Miguel de Navascués

2025-10-30

RESEARCH

I am a researcher at the CBGP (*Centre de Biologie pour la Gestion des Populations* - INRAE) in Montpellier. My work focuses on the development and evaluation of statistical methods for analyzing population genetic data to infer evolutionary processes, including genetic drift, gene flow, and selection. Currently, my research is centered on three lines:

1. Simulation-based methods for the **joint inference of demographic and adaptive processes**.
2. **Analysis of temporal population genetic data**, such as data from monitored or experimental populations.
3. **Integrative demographic inference** combining genetic and demographic data (such as capture-recapture data).

Career

2009–present **Researcher**, INRAE UMR1062 Centre de Biologie pour la Gestion des Populations (Montpellier, France).
2021–2023 **Visiting Researcher**, Human Evolution, EBC, Uppsala University (Sweden).
2019–2021 **Marie Skłodowska-Curie Fellow**, Human Evolution, EBC, Uppsala University (Sweden).
2007–2009 **Postdoctoral researcher**, CNRS UMR7625 Écologie et Évolution (Paris, France).
2006–2007 **Visiting researcher**, Université Libre de Bruxelles (Belgium).
2006 **Postdoctoral researcher**, Universidad Politécnica de Madrid & Centro de Investigación Forestal-INIA (Spain).
2001–2005 **Doctoral candidate**, University of East Anglia (UK).

Projects

2025–2029 **OPTIMISTII - Optimisation du contrôle de *Drosophila suzukii* via une synergie entre Technique de l'Insecte Stérile et Technique de l'Insecte Incompatible**. PARSADA, FranceAgriMer. Rol: Collaborator. PI: N. Rode.

2022–2025 DevOCGen: Développement et applications de nouveaux outils pour la gestion et la conservation des populations naturelles à partir de données génomiques. BiodivOc - Biodiversité Occitanie. Rol: Collaborator. PI: S. Boitard & R Leblois.

2022–2024 MODEGRAD: Modélisation de l'effet de la sélection naturelle, du changement climatique et de la gestion forestière sur la distribution des génotypes dans des gradients environnementaux. AMI CLIMAE 2021, INRAE. Rol: Co-coordinator. PI: I. Scotti.

2020–2025 INTROSPEC: Genomic consequences and evolutionary causes of introgression in the late stages of speciation. ANR AAPG 2019. Rol: Collaborator. PI: P.-A. Crochet.

2020–2025 PemilADAPT: Adaptive introgression in pearl millet. ANR AAPG 2019. Rol: Collaborator. PI: C. Berthouly-Salazar.

2019–2021 TimeAdapt: Tracking genetic adaptation of populations using time-series genomic data. Marie Skłodowska-Curie Individual Fellowships 2017. doi:10.5281/zenodo.3267084. Role: Fellow.

2018–2019 ABCSelection: Detection of loci under selection from temporal population genomic data through ABC random forest. LabEx Agro, Cemeb & Numev Joint Call 2016. Rol: PI.

2013–2016 SelfAdapt: Adaptation to climate change under selfing — experimental test and selection footprints in *Medicago truncatula*. Meta-programme INRA - Adapting Agriculture and Forestry to Climate Change 2012 (ACCAF). Rol: Co-coordinator. PI: L. Gay.

2012–2017 IBC: Institute de Biologie Computationelle. Investissements d'Avenir, Bioinformatique 2011. Rol: Collaborator. PI: O. Gascuel.

2014–2015 Interaction between ecological processes within community and population genetic structure. Programme PHC Tournesol. Rol: Host.

2010–2013 IGGiPop: Inférence en Génétique et Génomique des Populations. Jeune Équipe INRA 2010. Rol: Collaborator. PI: R. Vitalis.

2008–2009 Diversidad Genética Neutral y Adaptativa de *Taxus baccata* en la Red de Parques Nacionales. Ministerio de Medio Ambiente, Proyecto Parques Nacionales. Rol: Collaborator. PI: S. González-Martínez.

2007–2009 TRANSBIODIV: Biodiversité trans-spécifique neutre et fonctionnelle : développements théoriques et quantification chez des organismes modèles. ANR Biodiversité 2006. Rol: Postdoc. PI: M.-L. Cariou.

2006–2007 Developing Statistical Tools for Detection of Historical Demographic Expansions with Linked Microsatellites. European Science Foundation, CONGEN Exchange Grant — 1142. Rol: Grantee.

2005–2008 Introgresión genética en las poblaciones relictas de *Pinus sylvestris* L. var *nevadensis* del Parque Nacional de Sierra Nevada. Ministerio de Medio Ambiente, Proyecto Parques Nacionales. Rol: Postdoc. PI: J.J. Robledo-Arnuncio.

2003 Field Collection of Weevils (*Brachyderes* and *Rhinusa*) in Spain for Phylogeographic and Population Studies. Richard Warn Memorial Fund Travel Award 2003. Rol: Grantee.

Publications

citation statistics

```
library(scholar)
profileMNavascues <- get_profile("I9Rd7ocAAAAJ")
publicationsMNavascues <- get_publications("I9Rd7ocAAAAJ")
citation_historyMNavascues <- get_citation_history("I9Rd7ocAAAAJ")
par(mar=c(5,5,0,0)+0.2)
barplot(citation_historyMNavascues$cites,names.arg=citation_historyMNavascues$year,col="orange",b
box()
```

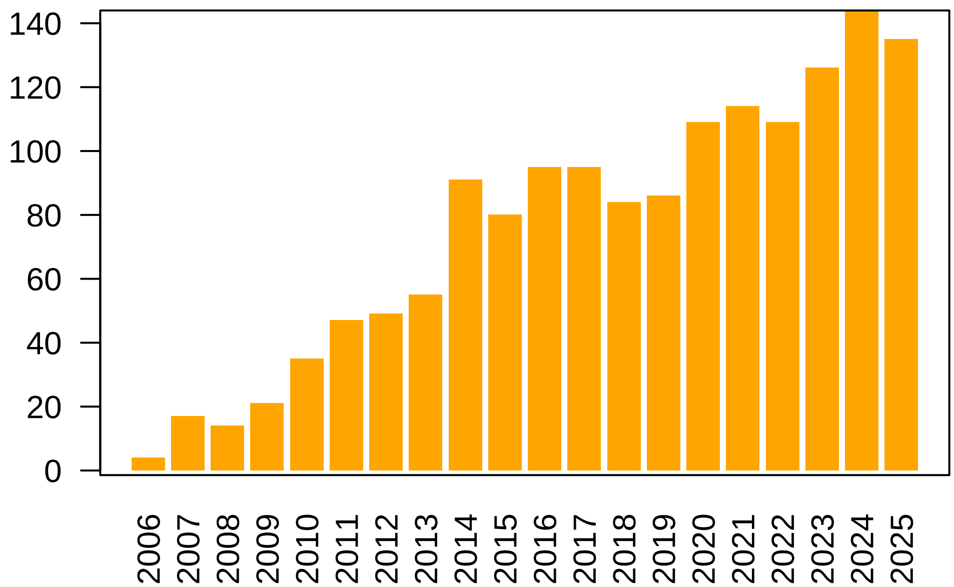

Figure 1: Number of citations per year according to Google Scholar.

preprints

1. **Retracing the center of origin and evolutionary history of nutmeg *Myristica fragrans*, an emblematic spice tree species**. J. Kusuma, N. Scarcelli, *M. de Navascués*, M. Couderc, P. Gerard, J. Duminil (2023) doi:10.22541/au.169114679.94713644/v1. Citations: 4.

2. **Revisiting the African mtDNA landscape: A continental update from complete mitochondrial genomes**. I. Lankheet, A. Chowdhury, C. Tellgren-Roth, C. Jolly, A.E.R. Soares, *M. de Navascués*, S. Pacchiarotti, L. Maselli, G. Kouarata, J.-P. Donzo, V. Coetzee, M. de Castro, P. Ebbesen, E. Priehodová, E. Podgorná, V. Černý, S.T. Green, P. Harena, L. Bakrobena, F. Leypey Mathew Fomine, Z. Gebre, M. Tolesa, W. Abebe Mengesh, M. de Jongh, H. Soodyall, K. Bostoen, C. Barbieri, M. Larena, H. Malmström, C.M. Schlebusch (2025) doi:10.1101/2025.04.05.647361. Citations: 1.

articles

1. **SelNeTime: a python package inferring effective population size and selection intensity from genomic time series data**. M. Uhl, P. Bunel, *M de Navascués*, S. Boitard, B. Servin (2024) doi:10.1101/2024.11.06.622284. Open Peer-Review, Open Code, Open Code. Citations: 0.
2. **Variation in the number of radiocarbon (14C) dates does not equal change in stone age population size over time**. A. Högberg, K. Brink, T. Brorsson, H. Malmström, E. Rudebeck, *M. de Navascués* (2025) doi:10.1016/j.jasrep.2025.105436. Open Data, Open Code. Citations: 0.
3. **Performance evaluation of adaptive introgression classification methods**. J. Romieu, G. Camarata, P.-A. Crochet, *M de Navascués*, R. Leblois, F. Rousset (2025) doi:10.24072/pcjournal.617. Open Peer-Review, Open Code. Citations: 3.
4. **Interplay between large low-recombining regions and pseudo-overdominance in a plant genome**. M. Salson, M. Duranton, S. Huynh, C. Mariac, C. Tranchant-Dubreuil, J. Orjuela, P. Cubry, A.-C. Thuillet, C. Burgarella, *M de Navascués*, L. Zekraouï, M. Couderc, S. Arribat, N. Rodde, A. Barnaud, A. Faye, N. Kane, Y. Vigouroux, C. Berthouly-Salazar (2025) doi:10.1038/s41467-025-61529-z. Open Peer-Review, Open Data, Open Code. Citations: 1.
5. **Analysis of the abundance of radiocarbon samples as count data**. *M. de Navascués*, C. Burgarella, M. Jakobsson (2024) doi:10.24072/pcjournal.522. Open Peer-Review, Open Code. Citations: 1.
6. **Mating systems and recombination landscape strongly shape genetic diversity and selection in wheat relatives**. C. Burgarella, M.-F. Brémaud, G. Von Hirschheydt, V. Viader, M. Ardisson, S. Santoni, V. Ranwez, *M. de Navascués*, J. David, S. Glémin (2024) doi:10.1093/evlett/qrae039. Open Code. Citations: 9.
7. **Genome scan of landrace populations of the self-fertilizing crop species rice, collected across time, revealed climate changes' selective footprints in the genes network regulating flowering time**. N. Ahmadi, M. Barry, J. Frouin, *M. de Navascués*, M. Touré (2023) doi:10.1186/s12284-023-00633-4. Citations: 3.
8. **Joint inference of adaptive and demographic history from temporal population genomic data**. V.A.C. Pavinato, S. De Mita, J.-M. Marin, *M. de Navascués* (2022) doi:10.24072/pcjournal.203. Open Peer-Review, Open Data, Open Code. Citations: 5.
9. **Ancient and historical DNA in conservation policy**. E.L. Jensen, D. Díez-del-Molino, M.T.P. Gilbert, L. Bertola, F. Borges, V. Cubric-Curik, *M. de Navascués*, P. Frandsen, M. Heuertz, C. Hvilsom, B. Jiménez-Mena, A. Miettinen, M. Moest, P. Pečnerová, I. Barnes, C. Vernesi (2022) {doi:10.1016/j.tree.2021.12.010. Citations: 61.
10. **Evolution of flowering time in a selfing annual plant: Roles of adaptation and genetic drift** L. Gay, J. Dhinaut, M. Jullien, R. Vitalis, *M de Navascués*, V. Ranwez, J. Ronfort (2022) doi:10.1002/ece3.8555. Open Peer-Review, Open Code. Citations: 6.
11. **Power and limits of selection genome scans on temporal data from a selfing population**. *M. de Navascués*, A. Becheler, L. Gay, J. Ronfort, K. Loridon, R. Vitalis (2021) doi:10.24072/pcjournal.47. Open Peer-Review, Open Code, Open Data. Citations: 7.

12. **Linking seascape with landscape genetics: oceanic currents favour colonisation across the Galápagos Islands by a coastal plant**. Y. Arjona, J. Fernández-López, *M. de Navascués*, N. Álvarez, M. Nogales, P. Vargas (2020) doi:10.1111/jbi.13967. Open Access. Citations: 13.
13. **Phylogeography of the striped field mouse, *Apodemus agrarius* (Rodentia: Muridae), throughout its distribution range in the Palearctic region**. A. Latinne, *M. de Navascués*, M. Pavlenko, I. Kartavtseva, R.G. Ulrich, M.-L. Tiouchichine, G. Catteau, H. Sakka, J.-P. Quéré, G. Chelomina, A. Bogdanov, M. Stanko, H. Lee, K. Neumann, H. Henttonen, J. Michaux (2020) doi:10.1007/s42991-019-00001-0. Open Access, Open Code. Citations: 27.
14. **Structure of multi-locus genetic diversity in predominantly selfing populations**. M. Jullien, *M. de Navascués*, J. Ronfort, K. Loridon, L. Gay (2019) doi:10.1038/s41437-019-0182-6. Open Data, Open Code. Citations: 32.
15. **Pleistocene origins of chorusing diversity in Mediterranean bush-cricket populations (*Ephippiger diurnus*)**. Y. Esquer-Garrigos, R. Streiff, V. Party, S. Nidelet, *M de Navascués*, M.D. Greenfield (2019) http://doi.org/10.1093/biolinnean/bly195. Open Data. Citations: 5.
16. **Discontinuities in quinoa biodiversity in the dry Andes: an 18-century perspective based on allelic genotyping**. T. Winkel, M.G. Aguirre, C.M. Arizio, C.A. Aschero, M.d.P. Babot, L. Benoit, C. Burgarella, S. Costa-Tártara, M.-P. Dubois, L. Gay, S. Hocsman, M. Jullien, S.M.L. Lopez Campeny, M.M. Manifesto, *M. de Navascués*, N. Oliszewski, E. Pintar, S. Zenboudji, H.D. Bertero, R. Joffre (2018) http://doi.org/10.1371/journal.pone.0207519. Open Code. Citations: 27.
17. **Intermediate degrees of synergistic pleiotropy drive adaptive evolution in ecological time**. L. Frachon, C. Libourel, R. Villoutreix, S. Carrere, C. Glorieux, C. Huard-Chauveau, *M de Navascués*, L. Gay, R. Vitalis, E. Baron, L. Amsellem, O. Bouchez, M. Vidal, V. Le Corre, D. Roby, J. Bergelson, F. Roux (2017) doi:10.1038/s41559-017-0297-1. Open Code. Citations: 121. ¡
18. **Identification of gene pools used in restoration and conservation by chloroplast microsatellite markers in Iberian pine species**. E. Hernández-Tecles, J. de las Heras, Z. Lorenzo, *M. de Navascués*, R. Alía (2017) doi:10.5424/fs/2017262-9030. Open Code. Citations: 2.
19. **Demographic inference through approximate-Bayesian-computation skyline plots**. *M. de Navascués*, R. Leblois, C. Burgarella (2017) doi:10.7717/peerj.3530. Open Code. Citations: 19.
20. **Increased fire frequency promotes stronger spatial genetic structure and natural selection at regional and local scales in *Pinus pinaster* Mill**. K.B. Budde, *M. de Navascués*, S.C. González-Martínez, C. Burgarella, E. Mosca, Z. Lorenzo, M. Zabal-Aguirre, G.G. Vendramin, M. Verdú, J.G. Pausas, M. Heuertz (2017) doi:10.1093/aob/mcw286. Citations: 38.
21. **Black rat invasion of inland Sahel: insights from interviews and population genetics in South Western Niger**. K. Berthier, M. Garba, R. Leblois, *M. de Navascués*, C. Tatard, P. Gauthier, S. Gagaré, S. Piry, C. Brouat, A. Dalecky, A. Loiseau, G. Dobigny (2016) doi:10.1111/bij.12836. Citations: 31.
22. **Genetic, morphological and acoustic evidence reveals lack of diversification in the colonization process in an island bird**. J.C. Illera, A.M. Palmero, P. Laiolo, F. Rodríguez, Á.C. Moreno, *M. de Navascués* (2014) doi:10.1111/evo.12429. Open Data. Citations: 32.

23. **Distinguishing migration from isolation using genes with intragenic recombination: detecting introgression in the *Drosophila simulans* species complex**. *M. de Navascués*, D. Legrand, C. Campagne, M.-L. Cariou, F. Depaulis (2014) doi:10.1186/1471-2148-14-89. Open Data. Citations: 9.

24. **The complex history of the olive tree: from Late Quaternary diversification of Mediterranean lineages to primary domestication in the northern Levant**. G. Besnard, B. Khadari, *M. de Navascués*, M. Fernández-Mazuecos, A. El Bakkali, N. Arrigo, D. Baali-Cherif, V. Brunini-Bronzini de Caraffa, S. Santoni, P. Vargas, V. Savolainen (2013) doi:10.1098/rspb.2012.2833. Citations: 324.

25. **Recent population decline and selection shape diversity of taxol-related genes**. C. Burgarella, *M. de Navascués*, M. Zabal-Aguirre, E. Berganzo, M. Riba, M. Mayol, G.G. Vendramin, S.C. González-Martínez (2012) doi:10.1111/j.1365-294X.2012.05532.x. Open Data. Citations: 34.

26. **Evolution of neutral and flowering genes along pearl millet (*Pennisetum glaucum*) domestication**. G. Lakis, *M. de Navascués*, S. Sekima, M. Simon, M.-S. Remigereau, M. Leveugle, N. Takvorian, F. Lamy, F. Depaulis, T. Robert (2012) doi:10.1371/journal.pone.0036642. Citations: 16.

27. **Mutation rate estimates for 110 Y-chromosome STRs combining population and father-son pair data**. C. Burgarella, *M. de Navascués* (2011) doi:10.1038/ejhg.2010.154. Citations: 113.

28. **CpDNA-based species identification and phylogeography: application to African tropical tree species**. J. Duminil, M. Heuertz, J.-L. Doucet, N. Bourland, C. Cruaud, F. Gavory, C. Doumenge, *M. de Navascués*, O.J. Hardy (2010) doi:10.1111/j.1365-294X.2010.04917.x. Citations: 64.

29. **Combining contemporary and ancient DNA in population genetic and phylogeographic studies**. *M. de Navascués*, F. Depaulis, B.C. Emerson (2010) doi:10.1111/j.1755-0998.2010.02895.x. Open Access. Citations: 61.

30. **Genetic diversity and fitness in small populations of partially asexual, self-incompatible plants**. *M. de Navascués*, S. Stoeckel, S. Mariette (2010) doi:10.1038/hdy.2009.159. Open Code. Citations: 45.

31. **Estimating gametic introgression rates in a risk assessment context: a case study with Scots pine relicts**. J.J. Robledo-Arnuncio, *M. de Navascués*, S.C. González-Martínez, L. Gil (2009) http://doi.org/10.1038/hdy.2009.78. Citations: 30.

32. **Elevated substitution rate estimates from ancient DNA: model violation and bias of Bayesian methods**. *M. de Navascués*, B.C. Emerson (2009) doi:10.1111/j.1365-294X.2009.04333.x. Open Data. Citations: 79.

33. **Characterization of demographic expansions from pairwise comparisons of linked microsatellite haplotypes**. *M. de Navascués*, O.J. Hardy, C. Burgarella (2009) doi:10.1534/genetics.108.098194. Open Code. Citations: 15.

34. **The effect of altitude on patterns of gene flow in the endemic Canary Island pine, *Pinus canariensis***. *M. de Navascués*, G.G. Vendramin, B.C. Emerson (2008) doi:10.1515/sg-2008-0052. Open Data. Citations: 18.

35. **Natural recovery of genetic diversity in reforested areas of the endemic Canary Island pine, *Pinus canariensis***. *M. de Navascués*, B.C. Emerson (2007) doi:10.1016/j.foreco.2007.04.009. Open Access, Open Data. Citations: 40.
36. **Narrow genetic base in forest restoration with holm oak (*Quercus ilex* L.) in Sicily**. C. Burgarella, *M. de Navascués*, Á. Soto, Á. Lora, S. Fici (2007) doi:10.1051/forest:2007055. Citations: 37.
37. **Chloroplast microsatellites reveal colonization and metapopulation dynamics in the Canary Island pine**. *M. de Navascués*, Z. Vaxevanidou, S.C. González-Martínez, J. Climent, L. Gil, B.C. Emerson (2006) doi:10.1111/j.1365-294X.2006.02960.x. Open Data. Citations: 75.
38. **Chloroplast microsatellite: measures of genetic diversity and the effect of homoplasy**. *M. de Navascués*, B.C. Emerson (2005) doi:10.1111/j.1365-294X.2005.02504.x. Citations: 100.

book chapter

1. **Introgresión genética en las poblaciones relictas de *Pinus sylvestris* L. var. *nevadensis* del Parque Nacional de Sierra Nevada**. J.J. Robledo-Arnuncio, *M. de Navascués*, S.C. González-Martínez, L. Gil (2009) in L Ramírez, B. Asensio (eds.) *Proyectos de Investigación en Parques Nacionales: 2005–2008* Organismo Autónomo Parques Nacionales, Madrid.

theses

1. **Genetic diversity of the endemic Canary Island pine tree, *Pinus canariensis***. *M. de Navascués* (2005) doi:10.6084/m9.figshare.1108027.v1. PhD thesis. University of East Anglia, School of Biological Sciences. Norwich (UK). Supervisor: B.C. Emerson.
2. **Metapopulation dynamics as an explanation to the genetic structure of the desert locust populations: parameter exploration**. *M. de Navascués* (2001) doi:10.6084/m9.figshare.1108026.v1. BSc research project. University of East Anglia, School of Biological Sciences. Norwich (UK). Supervisor: K. Ibrahim.

conference proceedings

1. **Diversidad y flujo genético en plantaciones de *Pinus canariensis***. *M. de Navascués*, B.C. Emerson (2006) in *Congreso de los Recursos Forestales. IV Jornadas Forestales de la Macaronesia*, Breña Baja (Spain).
2. **Estudio de la variabilidad genética de repoblaciones de *Quercus ilex* subsp. *ballota* en Andalucía**. C. Burgarella, *M. de Navascués*, S- Fici, Á- Lora González (2005) in *Libro de Resúmenes, Conferencias y Ponencias del IV Congreso Forestal Español*, Sociedad Española de Ciencias Forestales, Zaragoza (Spain).
3. **El éxito evolutivo de los insectos**. *M. de Navascués* (1997) in A. Delgado Buscalioni, J. Díaz Díaz, M. Herráiz Moreno (eds.) *I Encuentro de Estudiantes sobre Biología Evolutiva*, Universidad Autónoma de Madrid, Madrid (Spain).

Software and Data

Software

1. **DARth ABC: Analysis of the dated archaeological record through approximate Bayesian computation**. *M. de Navascués* (2025) doi:10.5281/zenodo.7560876.
2. **TimeAdapt: joint inference of demography and selection from longitudinal population genomic data**. *M de Navascués* (2025) doi:10.5281/zenodo.8232572.
3. **TrackingSelection: Tracking selection using temporal population genomics data**. V.A.C. Pavinato, S. De Mita, J.M. Cridland, *M. de Navascués* (2022) doi:10.5281/zenodo.7244868.
4. **DriftTest: a computer program to detect selection from temporal genetic differentiation in partially selfing populations**. *M. de Navascués*, R. Vitalis (2020) doi:10.5281/zenodo.1194662.
5. **DriftTest-Evaluation: A suite of R scripts to evaluate the performance of DriftTest for detection of selection with temporal data**. A. Becheler, R. Vitalis, *M. de Navascués* (2020) doi:10.5281/zenodo.1194666
6. **microsatABC-IM: Approximate Bayesian computation inferences under the isolation with migration model from microsatellite data} _M de Navascués} (2017) doi:10.5281/zenodo.895905 %Code from Latinne _et al}. (2020) doi:10.1007/s42991-019-00001-0
7. **Code from Structure of multilocus genetic diversity in predominantly selfing populations} M Jullien, _M de Navascués}, J Ronfort, L Gay (2019) doi:10.15454/VYPXIJ %Code from Jullien _et al}. (2019) doi:10.1038/s41437-019-0182-6
8. **ABCQuinoa: Scripts in R for an an approximate Bayesian computation analysis of ancient Quinoa gene pools and their contribution to modern varieties in Antofagasta} _M de Navascués}, C Burgarella (2018) doi:10.5281/zenodo.1489308 %Code from Winkel _et al}. (2018) doi:10.1371/journal.pone.0207519
9. **TempoDiff: a computer program to detect selection from temporal genetic differentiation} R Vitalis, L Gay, _M de Navascués} (2017) doi:10.5281/zenodo.375600 %Code from Frachon _et al}.:(2017) doi:10.1038/s41559-017-0297-1
10. **Blue Caterpillar: assignment methods on chloroplast microsatellites for provenance identification of reforested stands or seedlots} _M de Navascués}, J de Navascués (2020) doi:10.5281/zenodo.4052295 %Code from Hernández-Tecles et al. (2017) doi:10.5424/fs/2017262-9030
11. **DIYABCskylineplot: A suite of R scripts to perform a Approximate Bayesian Computation Skyline Plot} _M de Navascués} (2017) doi:10.5281/zenodo.267182 %Code from Navascués _et al}. (2017) doi:10.7717/peerj.3530
12. **SIplusA: A population genetics forward simulator with self-incompatibility system and partial asexuality**. *M de Navascués* (2016) doi:10.5281/zenodo.595878.
13. **LMSE (LinkMicroSatelliteExpansion): Characterization of demographic expansions from pairwise comparisons of linked microsatellite haplotypes} _M de Navascués} (2020) doi:10.5281/zenodo.4053654 %Code from Navascués _et al}. (2009) doi:10.1534/genetics.108.098194